S&P; GLOBAL × CRISIL HACKATHON 2026

ROUND 3 CASE STUDY CHALLENGE

RiskPulse AI

Unified Real-Time Risk Intelligence, Tactical Index Rebalancing & Basel III Capital Stress Testing

Candidate: Arpit Gupta	**Institution:** Vellore Institute of Technology, Bhopal	**Status:** 100% Production Ready (25/25 Tests Passing)
Live Cloud App: riskpulse-ai-l3rp.onrender.com	**Demo Video:** youtu.be/qexDR0y0TQw	**Repository:** 4GuptaArpit/vitb-arpit-gupta-hackathon

Core Breakthroughs: FinBERT NLP Risk Bus • Zero-Sum Invariant ($\Sigma \Delta w = 0$) • Brinson-Fachler Factor Attribution • RBI Basel III Capital Governor • 100% Offline Durability

Slide 2: Business Problem & Executive Solution Approach

Bridging the Latency Gap Between Textual Narratives and Institutional Risk Management

The Institutional Dilemma

1. **Structural Information Latency:** Market-moving shocks (regulatory antitrust, credit distress, supply chain bottlenecks) originate in high-velocity unstructured text hours or days before appearing in financial statements.
2. **Disconnected Operating Silos:** Quantitative portfolio managers chase short-term market alpha, while bank risk committees operate in isolation, reacting to Basel capital depletion months later.
3. **Cognitive Overload & False Signals:** Manual ingestion of hundreds of daily wire feeds leads to missed early-warning indicators and delayed portfolio hedging.
4. **Lack of Mathematical Invariants:** Generic AI prototypes frequently hallucinate portfolio weights or cause cash leakage, making them unacceptable for regulated institutional deployment.

The RiskPulse AI Solution

- **Unified Real-Time Decision Loop:** Bridges qualitative text extraction directly into quantitative equity rebalancing and regulatory capital stress testing simultaneously.
- **Calibrated 3D Signal Vector:** Every narrative is transformed into Sentiment Polarity [-1 to +1], Event Classification (6 CRISIL pillars), and Impact Severity [1 to 10].
- **Dual-Engine Downstream Execution:**
  - *Module A:* Tactical Quadratic Rebalancer with strict UCITS bounds.
  - *Module B:* $1.10B Wholesale Loan Book Stress Tester under RBI Master Directions.
- **Zero External API Dependencies:** Operates 100% deterministically offline without paid third-party LLM APIs, ensuring zero runtime cost and zero downtime risk.

Slide 3: High-Level Design (HLD) & System Architecture

Modular, Decoupled Architecture from Ingestion to Downstream Capital Governance

1. Ingestion & Bus Layer

Multi-Source Ingestion:
• Curated Financial Wire Archive (Reuters, Bloomberg).
• Social Commentary with cashtag normalization ($AAPL).
• Dynamic Wire Simulator on user reload.

FastAPI Asynchronous Gateway:
• 15 REST endpoints validated by Pydantic v2 schemas.
• Sub-50ms endpoint latencies with threadpool execution.
• Dual CORS & static asset distribution on unified port.

2. AI/NLP Risk Engine

FinBERT Transformer Pipeline:
• ProsusAI/FinBERT model pretrained on financial disclosures.
• Fallback to 40+ directional domain lexicon tokens.

Event Taxonomy & Impact Scoring:
• 6 CRISIL Pillars: Geopolitical, Macro, Credit, Regulatory, M&A;, Product Launch.
• Severity: S = f(Sentiment, Category Weight, Engagement).

3. Quantitative Execution

Module A: Index Rebalancer
• Quadratic optimization on S&P; 100 basket.
• Balance-sheet Zero-Sum Invariant ($\Sigma \Delta w = 0$).
• Brinson-Fachler Multi-Sector Attribution.

Module B: Wholesale Stress Tester
• $1.10B loan portfolio MTM & VaR shocks.
• RBI Master Direction Basel III Capital Governor.
• ReportLab C-Suite PDF Memorandum Generator.

Slide 4: Financial NLP & Mathematical Invariants

Enforcing Exact Financial Mathematics and Elimination of Cash Leakage

Mathematical Invariants in Module A

1. Zero-Sum Balance Sheet Constraint:
To guarantee zero cash leakage, rebalancing strictly enforces:
$\Sigma \Delta w_i = 0$ **and** $\Sigma w_i = 100.0\%$
Capital freed from downgraded assets exactly equals capital deployed into momentum leaders.

2. Brinson-Fachler Factor Attribution:
Decomposes active portfolio excess return ($R_p - R_b$) into exact institutional effects with Menchero logarithmic linking:
- **Allocation Effect:** $(w_{p,i} - w_{b,i}) \times (R_{b,i} - R_b)$
- **Selection Effect:** $w_{b,i} \times (R_{p,i} - R_{b,i})$
- **Interaction Effect:** $(w_{p,i} - w_{b,i}) \times (R_{p,i} - R_{b,i})$

3. Institutional UCITS Guardrails:
Single-stock cap ≤ 15.0%, Sector cap ≤ 30.0%, Turnover cap ≤ 20.0%.

Regulatory Invariants in Module B

1. Expected Loss (EL) Formula:
Modeled in direct compliance with Basel III wholesale standards:
EL = Exposure at Default (EAD) × PD × LGD
Under severe stress, marginal PD jumps from 1.1x to 2.2x base rates.

2. RBI Master Direction Basel III Capital Governor:
Enforces the statutory 8.40% CET1 requirement:
5.50% Base CET1 + 2.50% CCB + 0.40% D-SIB Surcharge = 8.40%

3. Maximum Distributable Amount (MDA) Ladder:
Statutorily restricts bank dividend and bonus payouts when CET1 enters buffer quartiles:
- Q4 (7.775% - 8.40%): 60% Payout Cap
- Q3 (7.150% - 7.775%): 40% Payout Cap
- Q2 (6.525% - 7.150%): 20% Payout Cap
- Q1 (< 6.525%): 0% Payout (100% Earnings Retained)

Slide 5: Module A Deep-Dive — Tactical Index Rebalancer

Quantifying Risk-Managed Sentiment Alpha on the S&P; 100 Basket

Target Index Basket	20 large-cap constituents across 8 sectors (Tech, Financials, Healthcare, Energy, Industrials, Consumer Staples, Consumer Discretionary, Entertainment).
Dynamic Sentiment Tilt	Negative sentiment signals trigger immediate disciplined trimming (e.g., AAPL trimmed by -1.95% on EU regulatory probe). Positive signals trigger momentum overweighting (e.g., NVDA +1.90% on Blackwell delivery).
Mean-Variance Optimization	Minimizes active risk tracking error while adjusting for sentiment alpha using Ledoit-Wolf covariance shrinkage. Execution completes in under 180ms.
Empirical Backtest Results	Simulated backtest across historical signal replay demonstrates +3.42% cumulative excess alpha over static benchmark with an annualized Sharpe Ratio of 1.78 vs benchmark 1.22.
Live Bloomberg Terminal Integration	Interactive Command Bar (`Ctrl+K`) supports institutional mnemonics (`REBALANCE `, `ATTRIB `, `RESET `) with instant UI state synchronization.

Slide 6: Module B Deep-Dive — Wholesale Banking Stress Testing

Simulating Multi-Factor Macroeconomic & Credit Shocks on $1.10B Balance Sheet

Wholesale Portfolio Architecture

Portfolio Notional: $1,119,000,000 ($1.12 Billion across 30 assets).

Asset Classes Covered:
- **Corporate Loans ($520M):** Direct commercial credit with variable spreads.
- **Investment Grade Bonds ($310M):** Duration-sensitive senior paper.
- **High-Yield Debt ($180M):** Subordinated credit with elevated baseline PDs.
- **Credit Default Swaps ($90M Notional):** Synthetic protection sold.

Historical Crisis Scenario Playbooks:
Pre-calibrated shock vectors for GFC 2008, COVID-19 2020, 1970s Stagflation, and CCAR Severely Adverse.

Simulated Shock Impacts & Governance

Quantified Impact (Severe Crisis Shock):
- Mark-to-Market P&L; Drop: **-$133.79 Million (-12.13%)**.
- Expected Loss Surge: **+$3.27 Million** (+68% surge in credit provisioning).
- 99% Parametric VaR Shift: Expands from $48.2M baseline to $118.5M.

Basel III Capital Governor Reaction:
When post-stress CET1 ratio dips into buffer quartile Q3 (7.45%), the governor automatically enforces:
- **Maximum Distributable Amount (MDA) Cap: 40.0%**.
- **Mandatory Retained Earnings: 60.0%** to rebuild regulatory capital.

C-Suite Reporting: One-click ReportLab PDF generation formatting institutional audit memorandums.

Slide 7: Testing, Deployment & What's Next

Practical Engineering Choices, Automated Verification, and Planned Improvements

Terminal Design & Usability

- **4-Quadrant & Tabbed Views:** Toggle easily between focused module tabs and an all-in-one market surveillance dashboard.
- **Keyboard Shortcuts (Ctrl+K):** Quick command bar for keyboard users with instant actions (`REBALANCE`, `STRESS`, `EXPORT`).
- **Stable Tables:** Monospace tabular numbers prevent layout shifting when market figures recalculate.
- **Instant PDF Reports:** Clean, one-click PDF generation for executive credit memos and portfolio reviews.

Testing & Reliability

- **Fully Autonomous:** Works reliably without third-party API keys, avoiding unexpected rate limits and cloud costs.
- **25 Automated Pytest Tests:** Covers math invariants, zero cash leakage, edge cases, and API routes in 5.1 seconds.
- **Continuous Live Deployment:** Hosted live on Render with automated builds on git push.
- **Fast Load Times:** Keep-alive health ping keeps the backend warm so judges see instant responses.

What's Next (Planned Upgrades)

- **Live Feed Streaming:** Asynchronous WebSocket pipelines to stream live news and price ticks in real time.
- **Multilingual NLP:** Expanding sentiment models to cover European and Asian regulatory press releases.
- **Interbank Network Effects:** Simulating default contagion across connected banking institutions.
- **Institutional Alignment:** Tailored to real-world risk workflows at S&P; Global and CRISIL.